Title Page

Engineering with Computers – Original Article

Certified benchmarking of discrete IS 800:2007-compliant space-truss sizing: an exact MILP baseline and a GA–local-search hybrid with local-optimality certificates

Short title: Certified benchmarking of discrete IS 800 truss sizing

Authors: Daipayan Mandal[1,2,*] and R. S. Sonparote[2]
Affiliations: [1]Department of Civil Engineering, Kavikulguru Institute of Technology and Science, Ramtek, Maharashtra, India. [2]Department of Applied Mechanics, Visvesvaraya National Institute of Technology, Nagpur, Maharashtra, India.
Corresponding author: Daipayan Mandal; `[e-mail]`
ORCID IDs: D. Mandal: [ORCID]; R. S. Sonparote: [ORCID]

Funding. The authors received no specific funding for this work.

Author contributions. D. Mandal: conceptualisation, methodology, software, validation, formal analysis, visualisation, writing – original draft. R. S. Sonparote: supervision, methodology review, writing – review and editing.

Competing interests. The authors declare no competing interests.

Ethics approval, consent to participate, consent for publication. Not applicable.

Data availability. All benchmark definitions, the section catalogue file, the raw results of every run (JSON), the MILP certificates and the scripts that regenerate every table and figure are openly available at `https://github.com/mdaipayan/GA-MINLP-IS800-Truss-Optimization` (folders `src_v2`, `results_v2`, `figures_v2`) and archived on Zenodo, `https://doi.org/10.5281/zenodo.20508136` [update to the DOI of the v2 release].

Code availability. The Python code (analysis engine, exact MILP, all optimizers, statistics) is released under the licence stated in the repository above.

Use of AI-assisted technologies. The authors used Claude (Anthropic) to assist with code review, LaTeX formatting and language editing. The authors checked all content and take full responsibility for the work.

Certified benchmarking of discrete IS 800:2007-compliant space-truss sizing: an exact MILP baseline and a GA–local-search hybrid with local-optimality certificates

Abstract

Metaheuristic studies of discrete truss sizing usually report the lightest design they found, but they rarely show how far this design is from the true optimum, and they often compare methods with different numbers of structural analyses. This paper addresses both problems for steel space trusses designed to IS 800:2007 with the 72 equal angles of IS 808:2021. First, following earlier exact formulations for trusses and for Eurocode frames, we write the complete problem—tension capacity, the column-buckling curve, slenderness and displacement limits—as an exact mixed-integer linear program (MILP) and use it as a yardstick. Branch-and-bound certifies the global optimum of a 6-bar truss (156.82 kg) and of the 25-bar benchmark (981.22 kg), and gives a proved lower bound for the 72-bar benchmark. Second, we propose GA–DNS, a genetic algorithm followed by a deterministic neighbourhood search that moves on the leg–thickness grid of the section table and ends with a local-optimality certificate. GA–DNS, a variant without the GA phase, six metaheuristics (SA, GA, PSO, ACO, BB-BC, DE) and two deterministic methods are compared over 20 seeded runs with the same analysis budget and the same feasibility rule. GA–DNS reached the certified optimum in all 40 runs on the 6- and 25-bar trusses, and on the 72-bar truss it gave significantly lighter designs than all six metaheuristics. Three results limit these claims. The variant without the GA phase performed equally well with fewer analyses. A simple fully stressed design also found both certified optima. The exact solver left a gap of 60 % on the 72-bar truss after 90 min. These results show where exact methods are sufficient and where heuristics are still needed.

Keywords: Discrete truss optimization; Mixed-integer linear programming; Global optimality; Genetic algorithm; Local search; IS 800:2007; Benchmarking

Article Highlights

- Discrete IS 800 truss sizing is posed exactly as an MILP and solved to certified optima.
- A local search on the section-table grid returns locally certified designs.
- Eight heuristics are compared at equal analysis budgets against certified optima.

1 Introduction

Space trusses are pin-jointed three-dimensional structures that carry load mainly by axial forces. They are used for long-span roofs, towers and industrial structures. In practice each member is chosen from a catalogue of rolled sections, so the minimum-weight design problem is discrete. In India the sections are those of IS 808:2021 [3] and the members must satisfy IS 800:2007 [2]: tension yielding, column buckling with the IS 800 buckling curve, and slenderness limits. Stiffness and displacement limits add further constraints.

Two research lines address this problem. Population-based metaheuristics have been tested for three decades on the same small set of space trusses, in particular the 25-bar and 72-bar benchmarks [28, 4, 5, 27, 7, 31]. In parallel, exact mixed-integer methods have solved discrete truss and frame problems to proven global optimality [1, 29, 16, 23, 36]. The two lines rarely meet. Stolpe [33] pointed out that, as a result, heuristic designs are compared only with other heuristic designs, methods are compared with different numbers of structural analyses, and many benchmarks that are routinely solved by heuristics could be solved exactly. Section 2 reviews both lines in more detail.

The present paper takes this critique as its starting point. We study discrete sizing of space trusses under IS 800:2007 and ask three questions. Can the complete IS 800 problem be solved exactly, and up to what size?

How close do common metaheuristics come to the certified optimum when they all receive the same analysis budget? Can a heuristic return a design together with a verifiable statement about its quality, instead of only a weight?

1.1 Contributions

1. We extend the exact MILP approach of Rasmussen and Stolpe [29], Mela [23] and Van Mellaert et al. [36] to IS 800:2007 and the IS 808:2021 equal angles: tension, the IS 800 buckling curve, slenderness (including stress reversal between load cases) and displacement limits. To our knowledge this is the first certified global optimum for an IS 800 truss problem. We then use the certified optima, and a proved lower bound where the MILP does not finish, as the reference for judging the heuristics (Sections 4.1 and 7.1).

2. We propose GA–DNS: a genetic algorithm for exploration followed by a deterministic discrete neighbourhood search (DNS) that moves on the leg–thickness grid of the section table. DNS stops only when every lighter neighbouring design has been analysed and is infeasible, so each final design carries a local-optimality certificate (Section 4.2).

3. We compare GA–DNS with six metaheuristics and two deterministic methods under one protocol: the same analysis model, the same feasibility rule, the same analysis budget, 20 seeded runs, distances to certified optima, data profiles and corrected non-parametric tests. An ablation without the GA phase shows which part of the hybrid does the work (Sections 5 and 7).

4. All code, raw results and certificates are openly available.

We use "best found" for heuristic results and "optimal" only when a certificate exists. We also report where the proposed method is *not* better than simpler alternatives.

Section 2 reviews related work, Section 3 states the problem, Section 4 the methods, Section 5 the comparison protocol and Section 6 the benchmarks. Section 7 presents the results, Section 8 discusses them and Section 9 concludes.

2 Related work

2.1 Metaheuristics on benchmark trusses

Genetic algorithms with a penalty transformation were among the first methods applied to discrete structural sizing [28]. They were followed by ant colony optimization [4], particle swarm optimization [27], Big Bang–Big Crunch [5], simulated annealing [20], teaching–learning-based optimization [7] and many other population-based methods; Saka and Geem [30] review more than a hundred such applications to steel structures. Most of these studies use the same small benchmarks (10-, 25-, 72- and 120-bar trusses) and report the lightest design found. Three common weaknesses have been pointed out. First, the designs are compared only with other heuristic designs, so the distance from the true optimum is unknown [33]. Second, the methods are often compared with different numbers of structural analyses and with tuned parameters, which makes the ranking unreliable. Hasançebi et al. [13] compared seven metaheuristics on four real-size pin-jointed structures designed to ASD-AISC, on one common coding platform; simulated annealing and evolution strategies were the strongest, while the simple GA converged slowly. Third, many "new" algorithms differ from older ones mainly in their metaphor, not in their search mechanism [32]. Good practice therefore requires equal budgets, many independent runs and proper non-parametric statistics [8], and profiles that show performance as a function of budget [10, 24].

2.2 Exact methods for discrete truss and frame sizing

For linear-elastic trusses with a finite set of cross-sections, the discrete problem can be written as a mixed-integer program with binary section-choice variables and continuous displacements and forces. Achtziger and Stolpe [1] solved discrete truss topology problems to guaranteed global optimality by branch-and-bound

Table 1: Representative studies on discrete or catalogue-based truss and frame sizing. "Proof": global optimality is certified. "Equal NFE": all methods compared at the same number of structural analyses.

Study	Method	Structures	Proof	Equal NFE	Constraints / code
Rajeev and Krishnamoorthy [28]	GA	3-bar truss to 160-bar tower	–	–	penalty on violation
Camp and Bichon [4]	ACO	space trusses	–	–	stress, deflection; several load cases
Camp [5]	BB-BC	benchmark space trusses	–	–	stress, deflection
Hasançebi et al. [13]	7 metaheuristics	4 real-size pin-jointed	–	common platform	ASD-AISC
Degertekin and Hayalioglu [7]	TLBO	truss benchmarks	–	–	stress, displacement
Achtziger and Stolpe [1]	branch-and-bound	truss topology	✓	n/a	discrete areas
Rasmussen and Stolpe [29]	cut-and-branch	truss topology	✓	n/a	discrete areas
Mela [23]	MILP	truss topology	✓	n/a	member buckling
Van Mellaert et al. [36]	MILP vs. GA	steel frames	✓	–	Eurocode checks
This work	MILP + GA–DNS + 6 heuristics	space trusses	✓[a]	✓	IS 800:2007, IS 808:2021

[a] 6-bar and 25-bar certified; 72-bar with proved lower bound.

and published benchmark examples with certified solutions. Rasmussen and Stolpe [29] added valid cuts and parallel branch-and-bound. Kanno and Guo [16] formulated robust truss topology design with stress constraints as an MILP. Mela [23] showed that member buckling, which is difficult in continuous truss optimization, becomes simple in a mixed-variable formulation, because the buckling resistance of each candidate profile is a constant. Van Mellaert et al. [35] obtained global optima for statically determinate trusses with displacement, member and joint constraints, and Van Mellaert et al. [36] extended the MILP approach to discrete sizing of steel frames with Eurocode member checks, comparing it with a genetic algorithm. Stolpe [33] concluded that many benchmark problems solved by metaheuristics in the literature can be solved to global optimality by such methods. The limits of the exact approach are problem size (the number of binaries grows with groups $\times$ sections) and nonlinear behaviour that does not fit a linear model.

2.3 Hybrid and memetic algorithms

Combining a population-based search with a local improvement step is the idea of memetic algorithms [25]. Krasnogor and Smith [19] gave a design taxonomy and Neri and Cotta [26] a review of the field. In structural optimization, GA phases are often followed by gradient, pattern or local discrete searches. The usual purpose is faster convergence. A by-product that is seldom used is that a complete local search in a finite neighbourhood ends with a checkable statement: no neighbouring design is both feasible and lighter. GA–DNS in this paper is a memetic algorithm in this sense; its new element is that this certificate is reported for every run and compared with the exact global optimum.

2.4 Beyond linear elasticity

Where the structural response is nonlinear, for example with global stability or geometric nonlinearity, exact MILP formulations are no longer directly available [9, 23]. This is where heuristics remain most useful, and it is the motivation for measuring their quality carefully on problems where the exact answer is known.

2.5 Research gap

Table 1 summarises representative studies. Three gaps follow.

1. Heuristic results on space-truss benchmarks are rarely measured against a certified optimum or a proved lower bound, and rarely at equal analysis budgets.

2. Exact MILP methods have been applied with generic stress, displacement and buckling limits and with Eurocode checks, but not, to our knowledge, with IS 800:2007 and the IS 808 section catalogue, which govern steel design in India.

3. Heuristics return a weight but no statement about the quality of the design; a local-optimality certificate is a cheap way to provide one.

3 Problem formulation

3.1 Design variables and objective

The geometry and the connectivity are fixed. The n_m members are divided into n_g design groups; all members of group g use the same section. The design variable of group g is an integer index $k_g \in \{1, \dots, n_k\}$ into the catalogue of the $n_k = 72$ equal leg angles of IS 808:2021 (Table 9) [3], from $20 \times 20 \times 3$ to $200 \times 200 \times 25$, kept in the order of the standard (by leg size, then by thickness). The catalogue gives the area A_k and the minimum radius of gyration $r_k = r_v$ (about the minor principal axis v–v). In our copy of the standard the r_v values of the angles with legs of 100 mm and more are printed without the decimal point; we therefore checked every row with $r_v = \sqrt{I_{vv}/A}$, which agrees with the printed digits. The weight is

$$W(\mathbf{k}) = \rho \sum_{g=1}^{n_g} A_{k_g} L_g, \qquad L_g = \sum_{i \in g} L_i, \tag{1}$$

with $\rho = 7850$ kg/m^3. Self-weight is not applied as a load, as in the original benchmarks.

3.2 Structural analysis

Let $\mathbf{B} \in \mathbb{R}^{n_m \times n_f}$ be the compatibility matrix that maps the free nodal displacements $\mathbf{u}$ to member elongations, $\mathbf{e} = \mathbf{B}\mathbf{u}$. For load case l the linear-elastic response follows from

$$\mathbf{K}(\mathbf{k})\, \mathbf{u}_l = \mathbf{p}_l, \qquad \mathbf{K} = \mathbf{B}^{\mathsf{T}} \operatorname{diag}\!\left(\frac{EA_i}{L_i}\right) \mathbf{B}, \qquad N_{il} = \frac{EA_i}{L_i}\, \mathbf{b}_i^{\mathsf{T}} \mathbf{u}_l, \tag{2}$$

with $E = 200$ GPa. $\mathbf{K}$ is factorised by Cholesky decomposition. Before any optimization, each model is checked for kinematic stability: the smallest eigenvalue of $\mathbf{K}$ (all members with the same section) must be positive. This simple check prevents a singular model from silently returning zero member forces.

3.3 IS 800:2007 member checks

Members are treated as concentrically loaded pin-ended bars with effective length KL, $K = 1$. For section k the design capacities are

$$T_k = \frac{A_k f_y}{\gamma_{m0}}, \tag{3}$$

$$C_{ik} = A_k f_{cd}(\lambda_{ik}), \quad \lambda_{ik} = \frac{KL_i}{r_k}, \quad f_{cd} = \frac{f_y/\gamma_{m0}}{\phi + \sqrt{\phi^2 - \bar{\lambda}^2}} \le \frac{f_y}{\gamma_{m0}}, \tag{4}$$

where $\bar{\lambda} = \sqrt{f_y/f_{cc}}$, $f_{cc} = \pi^2 E/\lambda^2$, $\phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$ and $\alpha = 0.49$ (IS 800 Cl. 7.1.2.1 with Table 7; buckling class c for angles, Table 10), with $f_y = 250$ MPa and $\gamma_{m0} = 1.10$. Buckling is checked about the weakest axis, with $r_k = r_v$. A design is feasible if, for every member i and load case l,

$$-C_{ik_g} \le N_{il} \le T_{k_g}, \qquad \lambda_{ik_g} \le \begin{cases} 180 & \text{if } N_{il} < 0 \text{ in any load case,} \\ 400 & \text{otherwise,} \end{cases} \qquad |u_{dl}| \le \delta_{\lim} \;\; \forall d \in \mathcal{D}, \tag{5}$$

where $\mathcal{D}$ is the set of displacement-limited degrees of freedom. The slenderness limits follow IS 800 Table 3. The ratio of demand to capacity is reported as DCR.

4 Methods

4.1 Exact MILP

For fixed member i and section k, the quantities T_k, C_{ik} and λ_{ik} in Equations (3) to (5) are *constants*. The non-smooth buckling curve therefore does not enter the optimization as a function; it only fixes one number for

each member–section pair. Following the displacement-based formulation of Rasmussen and Stolpe [29] and Stolpe [33], we introduce binary variables x_{gk} (group g uses section k), the displacements $\mathbf{u}_l$, and elongation variables e_{ikl}, which carry the elongation of member i only for the selected section:

$$\min \rho \sum_g \sum_k A_k L_g \, x_{gk} \tag{6}$$

$$\text{s.t. } \textstyle\sum_k x_{gk} = 1 \qquad \forall g \tag{7}$$

$$\mathbf{b}_i^{\mathsf{T}} \mathbf{u}_l = \textstyle\sum_k e_{ikl} \qquad \forall i, l \tag{8}$$

$$\textstyle\sum_i \mathbf{b}_i \sum_k \frac{EA_k}{L_i} \, e_{ikl} = \mathbf{p}_l \qquad \forall l \tag{9}$$

$$-\frac{L_i C_{ik}}{EA_k} \, x_{gk} \le e_{ikl} \le \frac{L_i T_k}{EA_k} \, x_{gk} \qquad \forall i \in g,\ k,\ l \tag{10}$$

$$|u_{dl}| \le \delta_{\lim} \qquad \forall d \in \mathcal{D},\ l \tag{11}$$

$$x_{gk} \in \{0, 1\}.$$

If $x_{gk} = 0$, Equation (10) forces $e_{ikl} = 0$; if $x_{gk} = 1$, the member force is $N_{il} = (EA_k/L_i)e_{ikl}$ and Equation (10) is exactly the tension and buckling check. The same idea—one continuous force or elongation variable per candidate section—is used by Mela [23] and Van Mellaert et al. [36]; here it is applied to the IS 800 capacities. Slenderness is handled in the same way: if $\lambda_{ik} > 180$ we set $C_{ik} = 0$ (the member may not be in compression with that section), and if $\lambda_{ik} > 400$ for a member of group g we fix $x_{gk} = 0$. The model is exact, and it contains no big-M constants, which gives a tighter continuous relaxation than the usual big-M form. Given an incumbent weight W^{inc}, sections that are too heavy even when all other groups use the lightest section are removed. The MILP is solved with the branch-and-bound solver of HiGHS [15], warm-started with the best design found by any heuristic, a relative gap tolerance of 10^{-6} and one CPU core. Every returned design is re-analysed with the stiffness method of Equation (2).

4.2 Proposed GA–DNS hybrid

GA–DNS has two phases (Algorithm 1).

Phase 1 (exploration). An integer GA with 40 individuals, binary tournament selection, uniform crossover ($p_c = 0.9$), mutation rate $1/n_g$ (80 % ± 1 or ± 2 catalogue steps, 20 % random reset) and elitism uses 50 % of the analysis budget. The best feasible design and the three best distinct designs of the last population are passed to Phase 2.

Phase 2 (deterministic refinement with certificate). Each elite is first made feasible, if necessary, by a greedy repair that moves the group with the largest constraint ratio one position up in the catalogue (for displacement violations, the move with the best stiffness gain per kilogram is chosen). The DNS then performs a descent with *catalogue moves*. The equal angles form a grid of leg size and thickness. A catalogue move of a group replaces its section by one of its grid neighbours: the next thinner or the next thicker angle with the same leg, or the angle of the next smaller or the next larger leg with the nearest thickness. A group therefore has at most four moves. The neighbourhood is

$$\mathcal{N}(\mathbf{k}) = \{\mathbf{k}' : \mathbf{k}' \text{ differs from } \mathbf{k} \text{ by a catalogue move in one or in two groups, and } W(\mathbf{k}') < W(\mathbf{k})\}, \tag{12}$$

which contains at most $4n_g + 8n_g(n_g - 1)$ designs. The two-group moves include "swaps", in which one group becomes lighter and another heavier. The neighbours are analysed in order of increasing weight, and the first feasible one is accepted. When no neighbour is feasible, DNS stops. At that point every design in $\mathcal{N}(\mathbf{k})$ has been analysed and is infeasible, so the final design is *certified* to be a local minimum with respect to $\mathcal{N}$. The certificate is a finite, checkable statement; it is not a proof of global optimality. The neighbourhood is defined on the section table itself and does not depend on how the catalogue is indexed. With the remaining budget, an iterated local search [21] changes two random groups of the best certified design by up to three catalogue positions and runs DNS again; a new design is kept only if it is certified and lighter.

Algorithm 1 GA–DNS

Require: model, analysis budget B
Run the GA (Phase 1) until $0.5B$ analyses are used
$\mathcal{E} \leftarrow$ best feasible design and 3 best distinct designs of the last population
for each $\mathbf{k} \in \mathcal{E}$ **do**
    $\mathbf{k} \leftarrow \textsc{Repair}(\mathbf{k})$; $\mathbf{k} \leftarrow \mathrm{DNS}(\mathbf{k})$ ▷ certified local minimum
end for
$\mathbf{k}^\star \leftarrow$ lightest certified design
while analyses $< B$ **do** ▷ iterated local search
    $\mathbf{k}' \leftarrow \mathbf{k}^\star$ with two random groups changed by ≤ 3 positions
    $\mathbf{k}' \leftarrow \textsc{Repair}(\mathbf{k}')$; $\mathbf{k}' \leftarrow \mathrm{DNS}(\mathbf{k}')$
    if $\mathbf{k}'$ is certified and $W(\mathbf{k}') < W(\mathbf{k}^\star)$ **then** $\mathbf{k}^\star \leftarrow \mathbf{k}'$
end while
return $\mathbf{k}^\star$ and its certificate

function DNS($\mathbf{k}$)
    repeat
        analyse the designs in $\mathcal{N}(\mathbf{k})$, lightest first
        move to the first feasible one; if none exists, **return** $\mathbf{k}$ (certified)
    until budget exhausted
end function

Table 2: Method settings (identical for all trusses).

Method	Settings
SA	one group changed by ±1–3 steps; T from $0.2W_{\max}$ to $10^{-4}W_{\max}$, geometric
GA	population 40; tournament 2; uniform crossover 0.9; mutation $1/n_g$; elitism 1
PSO	30 particles; inertia $0.9 \to 0.4$; $c_1 = c_2 = 2$; $\lvert v\rvert \leq 10$
ACO	20 ants; $\alpha = 1$, $\beta = 0.5$ ($\eta = 1/A_k$); evaporation 0.2; rank-based (3) + elitist update
BB-BC	30 candidates; centre = 0.5 (centre of mass)+0.5 (best); spread $\propto 1/t$
DE	DE/rand/1/bin; population 30; $F = 0.6$; $CR = 0.9$
GA–DNS	GA as above for $0.5B$; 4 elites; DNS neighbourhood Equation (12); ILS kick 2 groups, ≤ 3 positions
DNS–ILS	as GA–DNS, but Phase 1 is only the random initial population (40 analyses)
FSD	start at mid-catalogue; at most 50 resizing cycles
SQP-R	SLSQP, 200 iterations, ftol = 10^{-9}; round up; repair
MILP	HiGHS 1.15, gap 10^{-6}, 1 thread, time limit 5400 s, warm start

4.3 Comparison methods

Six metaheuristics act on the same integer variables: simulated annealing (SA) [18], the same GA as in Phase 1 run for the full budget, particle swarm optimization (PSO) [17, 27], ant colony optimization (ACO) [11, 4], Big Bang–Big Crunch (BB-BC) [12, 5] and differential evolution (DE) [34]. Continuous positions (PSO, BB-BC, DE) are rounded to the nearest index before analysis. All of them use the catalogue in the order of the standard, in which the radius of gyration changes smoothly from one position to the next. To see which part of the hybrid does the work, we also run DNS–ILS, an ablation of GA–DNS in which Phase 1 is reduced to the random initial population (40 analyses), so that almost the whole budget goes to DNS and the iterated local search. Two deterministic methods are included: fully stressed design (FSD), which gives every group the lightest angle that passes the member checks for the current forces and cannot control displacements, and SQP-R, a continuous SLSQP solution [38] with radius of gyration from a fitted law $r = c\,A^b$, followed by rounding up to the lightest angle with at least that area and the repair step of GA–DNS. Table 2 lists all settings. The parameters are common textbook values and were not tuned for any benchmark.

Table 3: Benchmark definitions.

Truss	Nodes	Members	Groups	Load cases	Budget B	Displacement limit
6-bar	4	6	6	1	3000	none
25-bar	10	25	8	1	10 000	8.89 mm, all nodes
72-bar	20	72	16	2	20 000	6.35 mm, top nodes x, y

5 Comparison protocol

Same analysis budget. One analysis (NFE) is one solution of Equation (2) for all load cases of one design. Every heuristic stops after $B = 3000$, 10 000 and 20 000 analyses for the 6-, 25- and 72-bar trusses. Repeated analyses of the same design are counted.

Same constraint handling. All heuristics rank designs by the feasibility rule of Deb [6], written as one scalar: $f = W$ for feasible designs and $f = W_{\max} + 10^3 v$ otherwise, where $W_{\max}$ is the weight of the heaviest catalogue design and v is the sum of constraint violations (normalised ratios minus one). A feasible design is therefore always better than an infeasible one, and no penalty factor needs tuning. A run that ends without a feasible design is counted as a failure. All reported designs are re-checked by an independent code path (Section 7.5).

Runs and statistics. Each stochastic method is run with seeds 0–19. We report best, median, mean, standard deviation, the share of feasible runs, the share of runs that reach the certified (or best-known) weight W^*, and the median number of analyses used to find the final design. Because the runs of different methods are independent, GA–DNS is compared with each competitor by the two-sided Mann–Whitney U test [22] with Holm correction [14], and the effect size is the Vargha–Delaney A_{12} [37]. Convergence is plotted against analyses, not against time or iterations, and data profiles [24] show the share of all runs that reach W^* (or $1.01W^*$) as a function of the budget used.

Software. Python 3.13, NumPy, SciPy [38] and HiGHS [15], on a 2-core Linux virtual machine. Effort is reported in structural analyses; wall-clock times are given only for the MILP.

6 Benchmark trusses

Figure 1 and Table 3 summarise the three trusses.

6-bar tetrahedron. Four nodes, six members, each member its own group. The apex carries 50 kN in y and -100 kN in z. No displacement limit. This small case is used to check the implementation.

25-bar transmission tower. The ASCE 25-bar tower with 10 nodes, 25 members in 8 groups and supports at the four base nodes, with the geometry and load case used by Camp and Bichon [4]: node 1: $(1, -10, -10)$ kip; node 2: $(0, -10, -10)$ kip; nodes 3 and 6: 0.5 kip in x. All nodal displacements are limited to ± 0.35 in (8.89 mm).

72-bar four-storey truss. Twenty nodes, 72 members in 16 groups (per storey: verticals, face diagonals, horizontals, plan diagonals), plan 120×120 in, storey height 60 in, supports at the four base nodes, and two load cases [5]: LC1, one top node $(5, 5, -5)$ kip; LC2, all four top nodes $(0, 0, -5)$ kip. The x and y displacements of the top nodes are limited to ± 0.25 in (6.35 mm).

The benchmark geometries and loads are taken from the earlier literature (imperial units, US sections and allowable-stress limits), but the section catalogue and the member checks here are those of IS 808 and IS 800. Therefore the weights in this paper must not be compared directly with the published results.

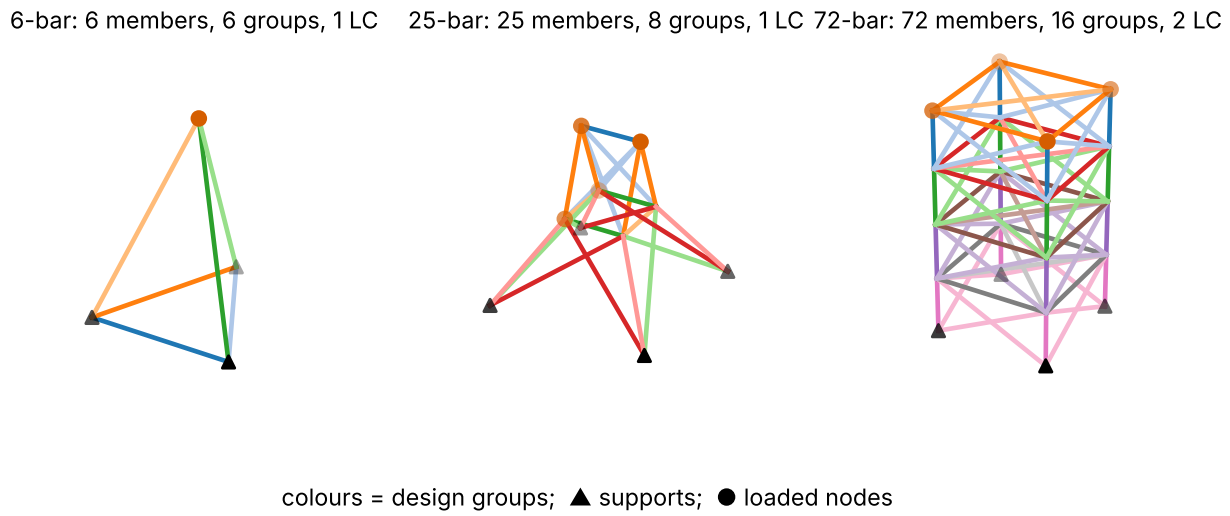


Figure 1: Benchmark trusses. Colours show design groups; triangles are supports and orange dots are loaded nodes.

Table 4: Exact MILP (Section 4.1) solved with HiGHS branch-and-bound. Lower bound = proved bound on the global optimum; best known = lightest feasible design found by any method; gap = (best known − lower bound)/best known. The 72-bar run was stopped at the 5400 s time limit. Times are wall-clock times on a machine shared with other jobs and are indicative only.

Truss	Binary	Continuous	Rows	B&B nodes	Time (s)	Lower bound (kg)	Best known (kg)	Gap (%)	Status
6-bar	432	438	576	0	0	156.82	156.82	0.0	optimal
25-bar	576	1,818	2,815	30,357	910	981.22	981.22	0.0	optimal
72-bar	1,152	10,464	17,120	2	60	657.31	1651.48	60.2	time limit

7 Results

7.1 Certified optima

Table 4 gives the exact MILP results. For the 6-bar truss the MILP proves at the root node that 156.82 kg is the global optimum. For the 25-bar truss it proves that 981.22 kg is the global optimum after 30 357 branch-and-bound nodes and about 15 min. For the 72-bar truss (1 152 binary variables) the solver reached the 90-minute limit with a proved lower bound of 657.31 kg, while the lightest feasible design found by any method weighs 1651.48 kg. The 72-bar optimum is therefore only known to lie in [657.31, 1651.48] kg, a gap of 60 %; we call 1651.48 kg the best-known value. This shows the practical limit of the exact approach with an open-source solver on a small machine: 16 groups × 72 sections and two load cases are already too difficult.

7.2 Statistical comparison

Table 5 lists the results of all methods.

- **6-bar.** GA–DNS and DNS–ILS reached the global optimum in all 20 runs. The six metaheuristics reached it in at most 40 % of their runs (ACO), and DE and BB-BC never did. FSD also finds the optimum, with only two analyses, because this truss is statically determinate and has no displacement limit.
- **25-bar.** GA–DNS and DNS–ILS again reached the certified optimum in all 20 runs. ACO and DE

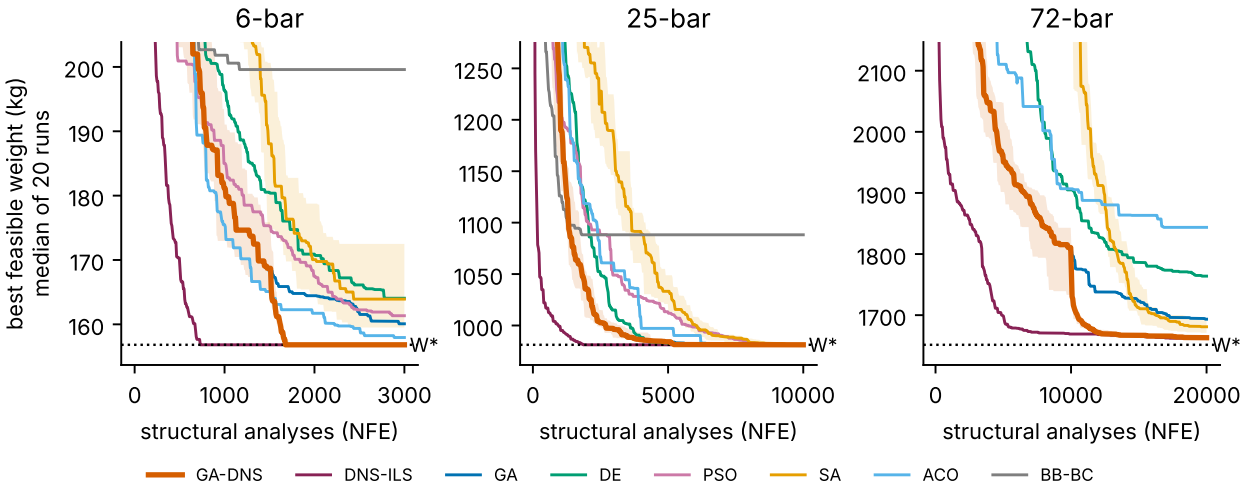


Figure 2: Median best-feasible weight of 20 runs against the number of structural analyses (shaded: interquartile range of GA–DNS and SA). Dotted line: certified optimum (6-, 25-bar) or best-known weight (72-bar).

reached it in 85 %, PSO in 75 %, GA in 65 % and SA in 45 % of the runs; PSO had some very poor runs (worst 2188 kg). FSD also returns the certified optimum after two analyses: the displacement limit is not active, and every group is governed by its own strength or slenderness check.

- **72-bar.** The best-known design (1651.48 kg) was found by one GA–DNS run; the best SA run reached 1652.00 kg and the best DNS–ILS run 1655.83 kg. The medians separate the methods more clearly: 1663.09 kg (GA–DNS) and 1661.75 kg (DNS–ILS) against 1681.01 kg (SA), 1693.40 kg (GA) and more than 1760 kg for DE, ACO, PSO and BB-BC. Sixteen GA–DNS runs and 14 DNS–ILS runs ended within 1 % of the best-known weight, against five SA runs and none of the other methods. BB-BC ended without a feasible design in three runs. FSD is 5.7 % heavier than the best-known design, so on this truss the redistribution of forces matters.

Table 6 gives the two-sided tests. GA–DNS is significantly lighter than all six metaheuristics on the 6-bar and the 72-bar truss ($A_{12} \geq 0.80$ and $A_{12} \geq 0.88$). On the 25-bar truss, where most methods often reach the optimum, the difference is significant against GA, SA and BB-BC but not against DE, PSO and ACO. In no case is a competitor significantly lighter than GA–DNS. GA–DNS and DNS–ILS do not differ significantly on any truss.

7.3 Convergence and data profiles

Figure 2 shows the median best-feasible weight against the number of analyses. The step of GA–DNS at $0.5B$ marks the start of DNS. DNS–ILS, which starts its local search after 40 analyses, converges much earlier: its median run reaches W^* after 724 analyses on the 6-bar truss and 1854 on the 25-bar truss, against 1673 and 5239 for GA–DNS and 5679 for the fastest metaheuristic on the 25-bar truss (DE). The data profiles in Figure 3 combine all 60 runs per method. With the full budget, GA–DNS and DNS–ILS reach the exact target W^* in 68 % and 67 % of all runs; the best metaheuristic is ACO with 42 %. For the 1 % target the shares are 93 % and 90 %, against 48 % or less for the metaheuristics. After a quarter of the budget, DNS–ILS has already reached the 1 % target in 60 % of all runs, and no other method in more than 5 %. The distributions of W/W^* are shown in Figure 4.

Table 5: Results over $N = 20$ independent runs (seeds 0–19) with an equal budget of structural analyses for every method. Statistics are over the feasible runs. SR = share of runs that end with an IS 800-feasible design; Hit = share of runs that reach W^* exactly; NFE = median number of analyses at which the final design was found. Bold = equal to W^*. FSD and SQP-R are deterministic (one run). *proposed; †ablation: the proposed method without the GA phase.

Method	Best (kg)	Median (kg)	Mean (kg)	Std (kg)	SR (%)	Hit (%)	NFE
6-bar — budget 3,000 analyses; $W^* = 156.82$ kg (certified global optimum)							
GA–DNS*	**156.82**	**156.82**	156.82	0.00	100	100	1,673
DNS–ILS†	**156.82**	**156.82**	156.82	0.00	100	100	724
GA	**156.82**	160.08	160.42	2.91	100	10	2,611
DE	157.66	164.06	164.64	4.81	100	0	2,754
PSO	**156.82**	161.34	160.83	3.70	100	20	2,810
SA	**156.82**	163.92	167.51	11.34	100	5	2,236
ACO	**156.82**	157.95	158.28	1.53	100	40	2,460
BB-BC	170.59	199.59	193.53	12.23	100	0	1,144
FSD	**156.82**	—	—	—	100	100	2
SQP-R	185.71	—	—	—	100	0	90
25-bar — budget 10,000 analyses; $W^* = 981.22$ kg (certified global optimum)							
GA–DNS*	**981.22**	**981.22**	981.22	0.00	100	100	5,239
DNS–ILS†	**981.22**	**981.22**	981.22	0.00	100	100	1,854
GA	**981.22**	**981.22**	981.75	1.15	100	65	7,876
DE	**981.22**	**981.22**	987.92	24.40	100	85	5,490
PSO	**981.22**	**981.22**	1136.23	382.77	100	75	8,216
SA	**981.22**	981.69	982.71	2.50	100	45	8,574
ACO	**981.22**	**981.22**	1001.32	49.42	100	85	5,260
BB-BC	1006.24	1088.19	1099.64	68.49	100	0	1,914
FSD	**981.22**	—	—	—	100	100	2
SQP-R	1019.55	—	—	—	100	0	152
72-bar — budget 20,000 analyses; $W^* = 1651.48$ kg (best known; not certified)							
GA–DNS*	**1651.48**	1663.09	1664.52	5.29	100	5	16,260
DNS–ILS†	1655.83	1661.75	1663.90	3.94	100	0	12,583
GA	1672.51	1693.40	1696.02	16.35	100	0	17,860
DE	1669.05	1763.87	1799.25	180.69	100	0	19,273
PSO	1682.44	3424.13	3166.34	1020.07	100	0	18,234
SA	1652.00	1681.01	1689.38	34.56	100	0	18,996
ACO	1674.06	1843.67	1975.89	322.36	100	0	14,368
BB-BC	2292.05	2678.20	2625.68	223.83	85	0	3,456
FSD	1746.16	—	—	—	100	0	6
SQP-R	1878.77	—	—	—	100	0	469

Table 6: Two-sided Mann–Whitney U tests of the final weights of GA–DNS against each other method, Holm-corrected within each truss, and Vargha–Delaney effect size A_{12} (probability that a GA–DNS run is lighter than a competitor run; 0.5 = no difference). Infeasible runs are ranked as worse than every feasible run.

Truss	Competitor	p_{Holm}	A_{12}	Conclusion ($\alpha = 0.05$)
6-bar	DNS–ILS	1.000	0.50	no significant difference
	GA	<0.001	0.95	GA–DNS lighter
	DE	<0.001	1.00	GA–DNS lighter
	PSO	<0.001	0.90	GA–DNS lighter
	SA	<0.001	0.97	GA–DNS lighter
	ACO	<0.001	0.80	GA–DNS lighter
	BB-BC	<0.001	1.00	GA–DNS lighter
25-bar	DNS–ILS	1.000	0.50	no significant difference
	GA	0.023	0.68	GA–DNS lighter
	DE	0.242	0.57	no significant difference
	PSO	0.079	0.62	no significant difference
	SA	<0.001	0.78	GA–DNS lighter
	ACO	0.242	0.57	no significant difference
	BB-BC	<0.001	1.00	GA–DNS lighter
72-bar	DNS–ILS	0.614	0.45	no significant difference
	GA	<0.001	0.99	GA–DNS lighter
	DE	<0.001	0.99	GA–DNS lighter
	PSO	<0.001	1.00	GA–DNS lighter
	SA	<0.001	0.88	GA–DNS lighter
	ACO	<0.001	1.00	GA–DNS lighter
	BB-BC	<0.001	1.00	GA–DNS lighter

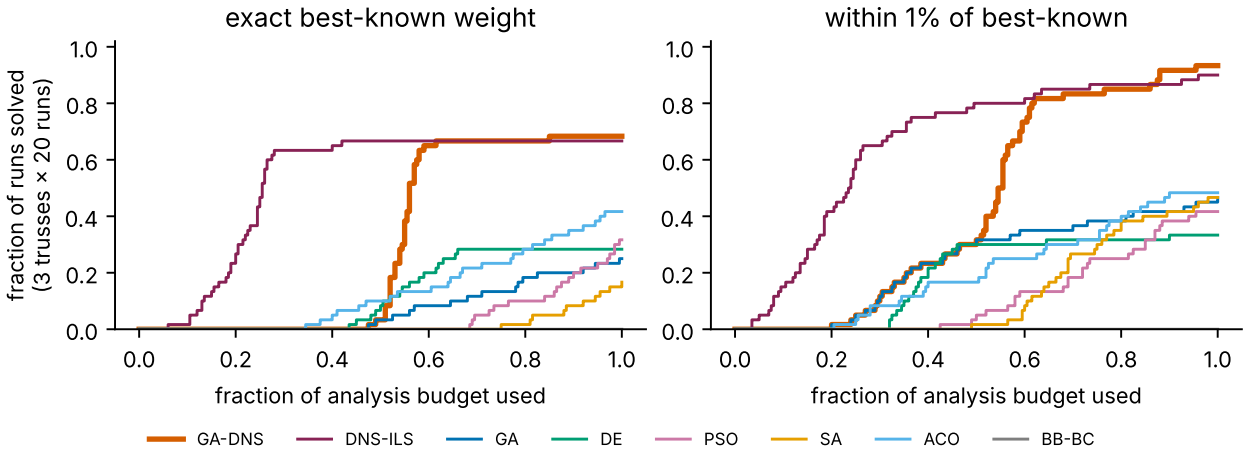


Figure 3: Data profiles over all three trusses (60 runs per method): share of runs that reach W^* exactly (left) or within 1 % (right) as a function of the fraction of the analysis budget used.

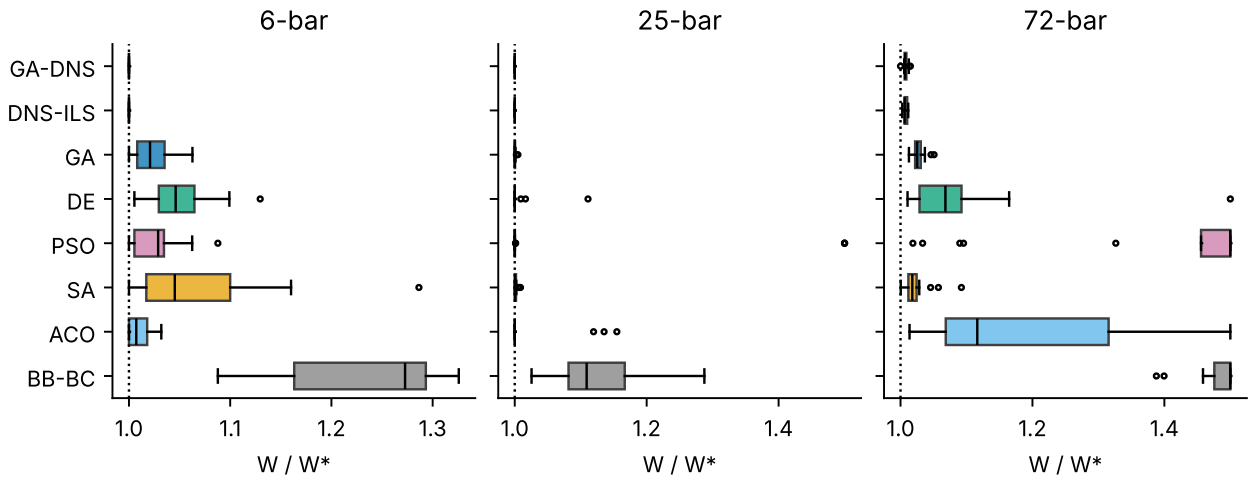


Figure 4: Distribution of the final weight ratio W/W^* over the feasible runs (1.0 = certified or best-known optimum; values above 1.5 are drawn at 1.5).

Table 7: Certified optimal 6-bar design (one member per group). Weight 156.82 kg; max. displacement ratio 0.000. F = range of member forces over all load cases (negative = compression); DCR = max. demand/capacity (yielding or buckling); λ-ratio = (KL/r_v)/limit.

Grp	n	Angle	A (cm^2)	r_v (mm)	$F_{\min}$ (kN)	$F_{\max}$ (kN)	KL/r_v	DCR	λ-ratio
1	1	90×90×6	10.60	18.0	-14.1	-14.1	167	0.267	0.926
2	1	45×45×3	2.67	9.0	24.5	24.5	373	0.403	0.932
3	1	45×45×3	2.67	9.0	24.5	24.5	373	0.403	0.932
4	1	60×60×4	4.73	12.0	9.4	9.4	377	0.088	0.943
5	1	60×60×4	4.73	12.0	9.4	9.4	377	0.088	0.943
6	1	130×130×10	25.10	25.7	-124.6	-124.6	166	0.994	0.923

7.4 Effect of the two phases and the certificates

Compared with the best design at the end of Phase 1, Phase 2 reduced the final weight of GA–DNS by a median of 7.0 % (6-bar), 0.3 % (25-bar) and 7.8 % (72-bar), with a maximum of 21.6 %, 1.5 % and 13.8 %. The ablation shows the opposite for Phase 1: without the GA, DNS–ILS reaches the same final quality (Table 6) with fewer analyses. On these benchmarks the local search on the section grid does the work, and the GA phase is not needed.

The final design was certified as a local minimum with respect to Equation (12) in 20, 20 and 16 of the 20 GA–DNS runs, and in 20, 20 and 15 of the DNS–ILS runs. In the remaining 72-bar runs the budget ended while the last check was still running; this includes the run that found the best-known design, which a separate check afterwards confirmed to be a local minimum. The certificate needed 57, 100 and 396 analyses for the best designs of the three trusses. On the 72-bar truss the certified local minima of different runs differ by about 1 % in weight, so a local certificate does not replace a global bound there.

7.5 Designs and independent verification

Tables 7 to 9 give the best designs. Each was re-analysed by a separate implementation (element-by-element assembly, dense solver, IS 800 formulas coded again). The weights agree, the equilibrium error is below 10^{-14} relative to the load, and all constraints are satisfied: the largest DCR is 0.994, 0.982 and 0.982, the largest slenderness ratio is 0.943, 0.991 and 0.980 of its limit, and the largest displacement ratio is 0.353 (25-bar) and 0.217 (72-bar). Thus, with the IS 800 slenderness limits and the minimum radius of gyration of single angles, strength and slenderness govern these designs, and the displacement limits are far from active.

Table 8: Certified optimal 25-bar design. Weight 981.22 kg; max. displacement ratio 0.353. F = range of member forces over all load cases (negative = compression); DCR = max. demand/capacity (yielding or buckling); λ-ratio = (KL/r_v)/limit.

Grp	n	Angle	A (cm^2)	r_v (mm)	$F_{\min}$ (kN)	$F_{\max}$ (kN)	KL/r_v	DCR	λ-ratio
1	1	25×25×3	1.45	4.9	5.2	5.2	389	0.157	0.972
2	4	100×100×6	11.80	20.0	-35.5	22.6	166	0.599	0.921
3	4	90×90×6	10.60	18.0	-61.2	24.3	151	0.982	0.837
4	2	25×25×3	1.45	4.9	1.2	2.1	389	0.064	0.972
5	2	55×55×4	4.33	11.0	-7.1	8.8	173	0.353	0.962
6	4	130×130×8	20.20	25.8	-21.0	7.9	178	0.235	0.991
7	4	130×130×8	20.20	25.8	-38.0	16.0	178	0.424	0.991
8	4	100×100×8	15.50	19.8	-69.8	36.3	171	0.948	0.951

Table 9: Best known 72-bar design (groups 1–4 = top storey; in each storey: verticals, face diagonals, horizontals, plan diagonals). Weight 1651.48 kg; max. displacement ratio 0.217. F = range of member forces over all load cases (negative = compression); DCR = max. demand/capacity (yielding or buckling); λ-ratio = (KL/r_v)/limit.

Grp	n	Angle	A (cm^2)	r_v (mm)	$F_{\min}$ (kN)	$F_{\max}$ (kN)	KL/r_v	DCR	λ-ratio
1	4	70×70×6	8.15	13.9	-20.6	-1.0	110	0.267	0.609
2	8	100×100×6	11.80	20.0	-11.2	5.6	170	0.199	0.947
3	4	90×90×6	10.60	18.0	-5.9	2.0	169	0.115	0.941
4	2	130×130×8	20.20	25.8	-8.9	1.3	167	0.089	0.928
5	4	50×50×3	2.99	10.0	-16.9	-0.9	152	0.982	0.847
6	8	100×100×6	11.80	20.0	-8.2	7.0	170	0.144	0.947
7	4	45×45×4	3.50	8.9	0.5	2.9	342	0.036	0.856
8	2	70×70×5	6.86	13.9	0.0	5.7	310	0.036	0.775
9	4	90×90×8	13.90	17.8	-23.4	11.7	86	0.132	0.476
10	8	100×100×6	11.80	20.0	-7.1	7.2	170	0.125	0.947
11	4	40×40×3	2.37	8.0	0.0	1.8	381	0.033	0.953
12	2	55×55×4	4.33	11.0	0.0	3.3	392	0.033	0.980
13	4	55×55×4	4.33	11.0	-27.2	19.8	139	0.933	0.770
14	8	100×100×6	11.80	20.0	-12.7	9.4	170	0.225	0.947
15	4	90×90×6	10.60	18.0	-2.5	4.5	169	0.049	0.941
16	2	55×55×4	4.33	11.0	0.2	1.6	392	0.016	0.980

8 Discussion

8.1 When is a heuristic justified?

For linear-elastic trusses the IS 800 problem is an exact MILP. On the 6-bar truss the MILP is solved at once. On the 25-bar truss it needs minutes, while the local search finds the same design within a few thousand analyses (about a second); however, only the MILP proves that this design is optimal. On the 72-bar truss the MILP could not close the gap in 90 minutes, so heuristics are still needed there, and better MILP techniques (valid cuts, symmetry breaking, parallel branch-and-bound [29]) or a commercial solver are an obvious next step. Our answer to the critique of Stolpe [33] is therefore: heuristic results on these benchmarks should be measured against exact bounds, and a heuristic is justified mainly where the exact model stops being an MILP—for example with geometric nonlinearity, imperfection-sensitive buckling or nonlinear code interaction checks—or where it becomes too large.

8.2 Simple methods first

Two simple methods did surprisingly well. FSD, which needs a handful of analyses, returned the certified optimum of the 6- and 25-bar trusses, because no displacement limit is active and each group is governed by its own member check. It was 5.7 % too heavy only on the 72-bar truss. DNS–ILS, a plain iterated local search without any population, matched GA–DNS everywhere. Population-based metaheuristics were less reliable than both on all three trusses. This agrees with the warning of Sörensen [32] and with the observation of Hasançebi et al. [13] that simple GAs converge slowly. We therefore recommend that studies of this kind always include FSD and a plain local search as baselines. SQP-R was 18.4 %, 3.9 % and 13.8 % heavier than W^*; the rounding step and the approximate $r(A)$ law cost weight.

8.3 Why the neighbourhood matters

For single angles the minimum radius of gyration depends almost only on the leg size ($r_v \approx 0.2$ times the leg), while the area depends on leg and thickness. Slenderness, which governs many members here, is therefore controlled by the leg, and strength by leg and thickness together. A neighbourhood that moves separately in leg and in thickness follows this structure. It is also the reason why all index-based methods use the catalogue in the order of the standard and not sorted by area: in area order, adjacent positions often belong to different legs, and the slenderness of a member jumps up and down from one position to the next.

8.4 What the certificate adds

A weight alone does not tell the engineer whether the design can still be improved. The DNS certificate is weaker than global optimality, but it is cheap (57 to 396 analyses here, and never more than $4n_g+8n_g(n_g-1)$), it can be verified by anyone, and it is independent of the random seed. Combined with the MILP lower bound, the user also gets a bound on the remaining gap: the best 72-bar design is certified locally optimal and lies at most 60 % above the global optimum. This bound is loose; it shows the value of a better lower bound rather than the quality of the design.

8.5 Limitations

- Only three small trusses were studied, and the 6-bar truss is trivial. Larger benchmarks are needed to see how the methods scale.
- The analysis is linear-elastic with ideal pin joints and concentric forces. Single angles connected through one leg should be checked with the equivalent slenderness of IS 800 Cl. 7.5, and net-section rupture and block shear (Cl. 6.3–6.4) are not checked. These checks would change the capacities but not the MILP structure, as long as each capacity is a constant for a member–section pair.
- The loads come from the imperial benchmarks of the earlier literature. They are not IS 875 load combinations, and the same loads are used for the strength checks and for the displacement limits.

- The method parameters were not tuned and no sensitivity study was made. Other settings could change the ranking of the metaheuristics, but they would not change the certified optima.
- Only one machine and one open-source MILP solver were used.

9 Conclusions

1. For linear-elastic space trusses with a fixed catalogue, the complete IS 800:2007 sizing problem (tension, column buckling, slenderness and displacements) can be written as an exact MILP, as earlier shown for other codes. With the 72 equal angles of IS 808:2021 it certified the global optima of the 6-bar (156.82 kg) and 25-bar (981.22 kg) trusses and gave a lower bound of 657.31 kg for the 72-bar truss (best known 1651.48 kg).
2. A deterministic neighbourhood search on the leg–thickness grid of the section table, restarted by an iterated local search, reached both certified optima in every run and gave significantly lighter 72-bar designs than SA, GA, PSO, ACO, BB-BC and DE at equal analysis budgets. The final design of GA–DNS carried a local-optimality certificate in 56 of 60 runs.
3. The GA phase of GA–DNS was not needed: the variant without it gave the same final weights with fewer analyses.
4. Fully stressed design found both certified optima with a handful of analyses and was 5.7 % too heavy on the 72-bar truss. Simple baselines of this kind should be reported together with metaheuristics.
5. With the IS 800 slenderness limits and the minimum radius of gyration of single angles, strength and slenderness, not displacement, govern the optimal designs of these benchmarks.
6. Future work will use the same certified protocol for larger trusses and for geometrically nonlinear and imperfection-sensitive space trusses, where an exact MILP is no longer available and heuristics with certificates are most needed.

References

[1] Achtziger W, Stolpe M (2007) Truss topology optimization with discrete design variables – guaranteed global optimality and benchmark examples. Struct Multidisc Optim 34(1):1–20. https://doi.org/10.1007/s00158-006-0074-2

[2] Bureau of Indian Standards (2007) IS 800:2007 General construction in steel – Code of practice (third revision). BIS, New Delhi

[3] Bureau of Indian Standards (2021) IS 808:2021 Hot rolled steel beam, column, channel and angle sections – Dimensions and properties (fourth revision). BIS, New Delhi

[4] Camp CV, Bichon BJ (2004) Design of space trusses using ant colony optimization. J Struct Eng 130(5):741–751. https://doi.org/10.1061/(ASCE)0733-9445(2004)130:5(741)

[5] Camp CV (2007) Design of space trusses using Big Bang–Big Crunch optimization. J Struct Eng 133(7):999–1008. https://doi.org/10.1061/(ASCE)0733-9445(2007)133:7(999)

[6] Deb K (2000) An efficient constraint handling method for genetic algorithms. Comput Methods Appl Mech Eng 186(2–4):311–338. https://doi.org/10.1016/S0045-7825(99)00389-8

[7] Degertekin SO, Hayalioglu MS (2013) Sizing truss structures using teaching-learning-based optimization. Comput Struct 119:177–188. https://doi.org/10.1016/j.compstruc.2012.12.011

[8] Derrac J, García S, Molina D, Herrera F (2011) A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. Swarm Evol Comput 1(1):3–18. https://doi.org/10.1016/j.swevo.2011.02.002

[9] Descamps B, Filomeno Coelho R (2014) The nominal force method for truss geometry and topology optimization incorporating stability considerations. Int J Solids Struct 51(13):2390–2399. https://doi.org/10.1016/j.ijsolstr.2014.03.003

[10] Dolan ED, Moré JJ (2002) Benchmarking optimization software with performance profiles. Math Program 91(2):201–213. https://doi.org/10.1007/s101070100263

[11] Dorigo M, Gambardella LM (1997) Ant colony system: a cooperative learning approach to the traveling salesman problem. IEEE Trans Evol Comput 1(1):53–66. https://doi.org/10.1109/4235.585892

[12] Erol OK, Eksin I (2006) A new optimization method: Big Bang–Big Crunch. Adv Eng Softw 37(2):106–111. https://doi.org/10.1016/j.advengsoft.2005.04.005

[13] Hasançebi O, Çarbaş S, Doğan E, Erdal F, Saka MP (2009) Performance evaluation of metaheuristic search techniques in the optimum design of real size pin jointed structures. Comput Struct 87(5–6):284–302. https://doi.org/10.1016/j.compstruc.2009.01.002

[14] Holm S (1979) A simple sequentially rejective multiple test procedure. Scand J Stat 6(2):65–70

[15] Huangfu Q, Hall JAJ (2018) Parallelizing the dual revised simplex method. Math Program Comput 10(1):119–142. https://doi.org/10.1007/s12532-017-0130-5

[16] Kanno Y, Guo X (2010) A mixed integer programming for robust truss topology optimization with stress constraints. Int J Numer Methods Eng 83(13):1675–1699. https://doi.org/10.1002/nme.2871

[17] Kennedy J, Eberhart R (1995) Particle swarm optimization. In: Proc ICNN'95, vol 4, pp 1942–1948. https://doi.org/10.1109/ICNN.1995.488968

[18] Kirkpatrick S, Gelatt CD, Vecchi MP (1983) Optimization by simulated annealing. Science 220(4598):671–680. https://doi.org/10.1126/science.220.4598.671

[19] Krasnogor N, Smith J (2005) A tutorial for competent memetic algorithms: model, taxonomy, and design issues. IEEE Trans Evol Comput 9(5):474–488. https://doi.org/10.1109/TEVC.2005.850260

[20] Lamberti L (2008) An efficient simulated annealing algorithm for design optimization of truss structures. Comput Struct 86(19–20):1936–1953. https://doi.org/10.1016/j.compstruc.2008.02.004

[21] Lourenço HR, Martin OC, Stützle T (2003) Iterated local search. In: Glover F, Kochenberger GA (eds) Handbook of Metaheuristics. Kluwer Academic Publishers, Boston, pp 320–353. https://doi.org/10.1007/0-306-48056-5_11

[22] Mann HB, Whitney DR (1947) On a test of whether one of two random variables is stochastically larger than the other. Ann Math Stat 18(1):50–60. https://doi.org/10.1214/aoms/1177730491

[23] Mela K (2014) Resolving issues with member buckling in truss topology optimization using a mixed variable approach. Struct Multidisc Optim 50(6):1037–1049. https://doi.org/10.1007/s00158-014-1095-x

[24] Moré JJ, Wild SM (2009) Benchmarking derivative-free optimization algorithms. SIAM J Optim 20(1):172–191. https://doi.org/10.1137/080724083

[25] Moscato P (1989) On evolution, search, optimization, genetic algorithms and martial arts: towards memetic algorithms. C3P Report 826, Caltech Concurrent Computation Program, California Institute of Technology, Pasadena

[26] Neri F, Cotta C (2012) Memetic algorithms and memetic computing optimization: a literature review. Swarm Evol Comput 2:1–14. https://doi.org/10.1016/j.swevo.2011.11.003

[27] Perez RE, Behdinan K (2007) Particle swarm approach for structural design optimization. Comput Struct 85(19–20):1579–1588. https://doi.org/10.1016/j.compstruc.2006.10.013

[28] Rajeev S, Krishnamoorthy CS (1992) Discrete optimization of structures using genetic algorithms. J Struct Eng 118(5):1233–1250. https://doi.org/10.1061/(ASCE)0733-9445(1992)118:5(1233)

[29] Rasmussen MH, Stolpe M (2008) Global optimization of discrete truss topology design problems using a parallel cut-and-branch method. Comput Struct 86(13–14):1527–1538. https://doi.org/10.1016/j.compstruc.2007.05.019

[30] Saka MP, Geem ZW (2013) Mathematical and metaheuristic applications in design optimization of steel frame structures: an extensive review. Math Probl Eng 2013:271031. https://doi.org/10.1155/2013/271031

[31] Shallan O, Eraky A, Sakr T, Hamdy O (2014) Optimization of plane and space trusses using genetic algorithms. Int J Eng Innov Technol 3(7):66–73

[32] Sörensen K (2015) Metaheuristics – the metaphor exposed. Int Trans Oper Res 22(1):3–18. https://doi.org/10.1111/itor.12001

[33] Stolpe M (2016) Truss optimization with discrete design variables: a critical review. Struct Multidisc Optim 53(2):349–374. https://doi.org/10.1007/s00158-015-1333-x

[34] Storn R, Price K (1997) Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. J Glob Optim 11(4):341–359. https://doi.org/10.1023/A:1008202821328

[35] Van Mellaert R, Lombaert G, Schevenels M (2016) Global size optimization of statically determinate trusses considering displacement, member, and joint constraints. J Struct Eng 142(2):04015120. https://doi.org/10.1061/(ASCE)ST.1943-541X.0001377

[36] Van Mellaert R, Mela K, Tiainen T, Heinisuo M, Lombaert G, Schevenels M (2018) Mixed-integer linear programming approach for global discrete sizing optimization of frame structures. Struct Multidisc Optim 57(2):579–593. https://doi.org/10.1007/s00158-017-1770-9

[37] Vargha A, Delaney HD (2000) A critique and improvement of the CL common language effect size statistics of McGraw and Wong. J Educ Behav Stat 25(2):101–132. https://doi.org/10.3102/10769986025002101

[38] Virtanen P, Gommers R, Oliphant TE, et al. (2020) SciPy 1.0: fundamental algorithms for scientific computing in Python. Nat Methods 17(3):261–272. https://doi.org/10.1038/s41592-019-0686-2